TUTORIAL – 10 | Automation & Regression Analysis

Aim: Analyze black-box testing, white-box testing, integration testing, and system testing on a sample web / mobile application.

Sample Application Chosen: Instagram (Mobile Application)

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1. Introduction

This practical applies Software Testing techniques — Black-box testing, Integration testing and System testing — to the Instagram mobile application. Instagram provides Login, Home Feed, Search/Explore, Profile, Direct Messages (DM), and Settings features. The objective is to review the visible user flows and interface using the supplied mobile screenshots and record expected and observed results.

2. Testing Techniques Applied

• Black-box Testing: Reviewed the visible Login screen, Home Feed, Search/Explore grid, Profile, Direct Messages and Settings without examining internal code.

• Integration Testing: Reviewed navigation between visible modules, including Home Feed → Search/Explore, Search → Profile, and Profile/Menu → Settings.

• System Testing: Considered the overall user journey across Login, browsing content, opening a profile, checking messages and opening Settings, based on the supplied screenshots.

• UI / Regression Checks: Checked visible screen layout, labels, icons, navigation and screen content in the supplied Android screenshots.

3. Test Environment

Item	Details
Application	Instagram Mobile App
Platform	Android / iOS (Mobile)
Network	Wi-Fi / Mobile Data
Tester	Roshni Rajput – Silver Oak University
Test Case ID	ST_10
Focus Areas	Login, Home Feed, Post View, Search/Explore, Profile, DM, Settings, UI

4. Test Scenarios, Steps & Screenshots

4.1 Login Module

Scenario 1a – Login with valid credentials

Test Steps:\n1. Open Instagram application\n2. Enter username, email address or mobile number\n3. Enter password\n4. Tap “Log in”

Expected Result: User is authenticated and redirected to the Home Feed.

Actual Result: The login screen is displayed with username/email/mobile and password fields, Log in, Forgotten password? and Create new account options. A successful login is not shown in this screenshot.

Status: PASS (screen display only)

Screenshot – Login Page:

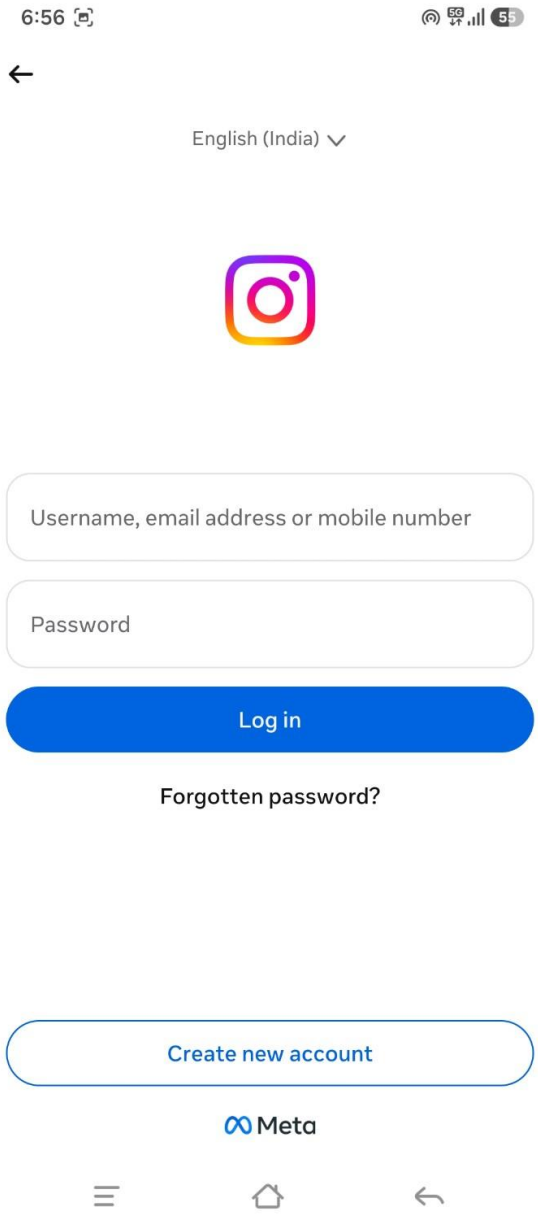


Figure 1: Instagram login screen with login fields and account recovery/creation options

Scenario 1b – Login with wrong password / empty fields / Forgot password

Test Steps: Observe the login form and locate the “Forgotten password?” and “Create new account” options.

Expected Result: Login and account recovery/creation options are visible and clearly labelled.\nActual Result: The options are visible on the supplied login screen. Password reset was not executed.\nStatus: PASS

4.2 Home Feed & Post Interaction

Scenario 2a – Home Feed loads after successful login

Test Steps: After login, observe the Home Feed for posts from followed accounts.

Expected Result: Posts appear with like, comment, share and profile options.

Actual Result: Home Feed displayed a post from silveroakuni with post media and engagement/navigation controls.

Status: PASS

Screenshot – Home Feed:

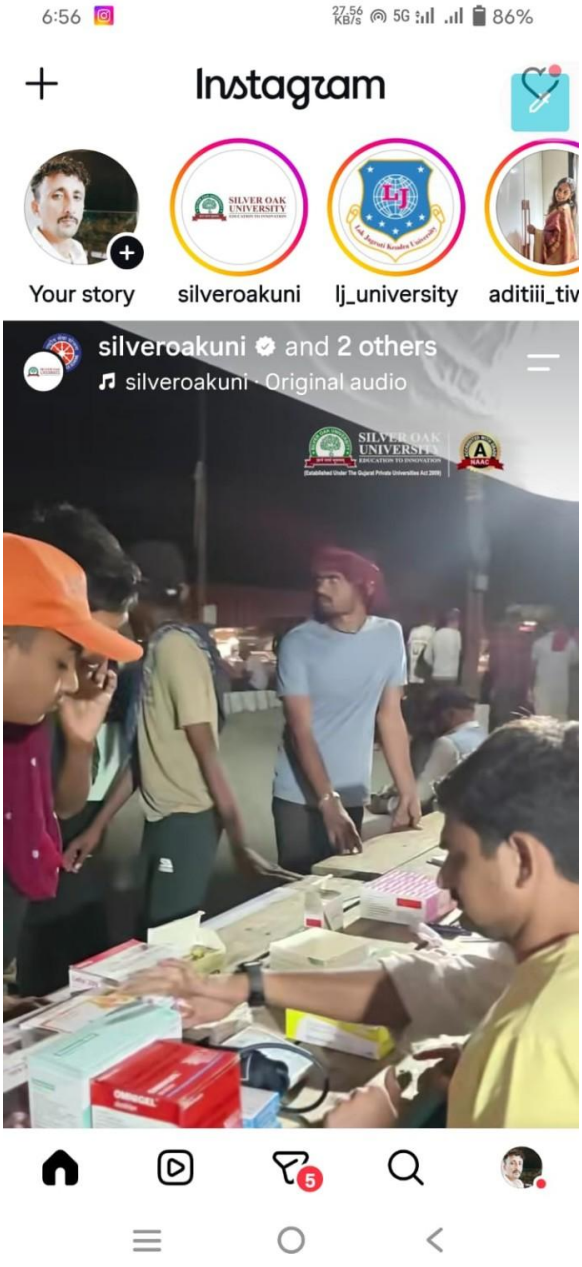


Figure 2: Instagram Home Feed showing a post from silveroakuni

Scenario 2b – Post detail view

Test Steps: Open any post → view full media, caption, engagement metrics.

Status: PASS

Screenshot – Single Post View:



Figure 3: Instagram post view from virat.kohli showing media and engagement controls

4.3 Search / Explore Module

Scenario 3a – Explore page grid loads

Test Steps: Tap Search (magnifying glass) icon → observe Explore grid of recommended posts/Reels.

Expected Result: Grid of posts with view counts appears.

Actual Result: Explore page loaded with multiple posts and view counts.

Status: PASS

Screenshot – Explore / Search Page:

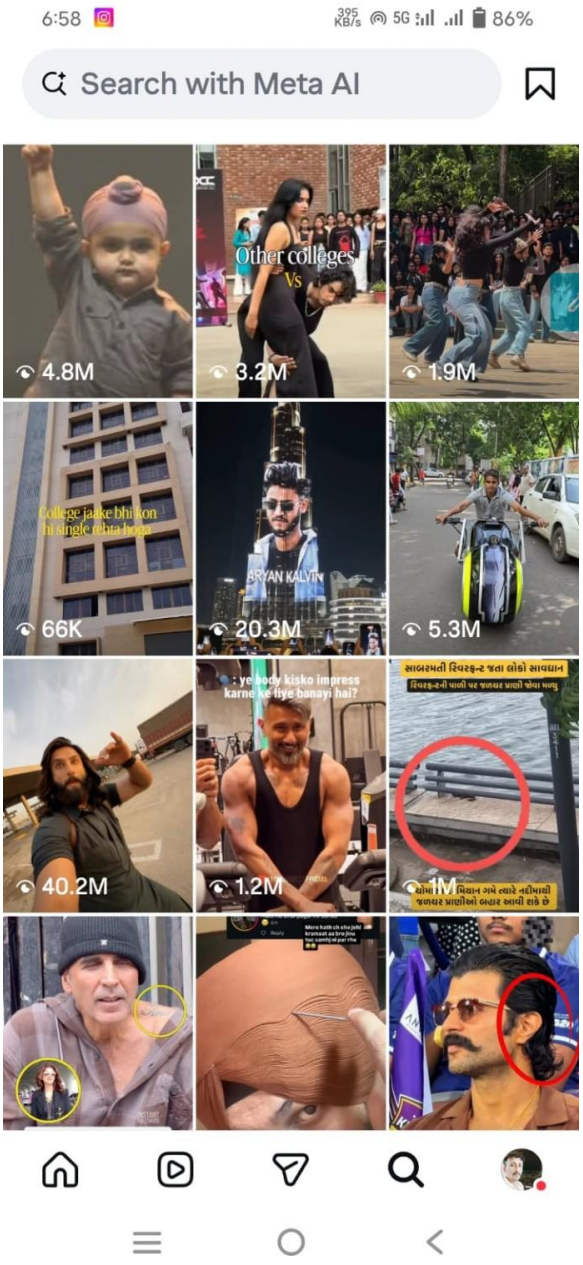


Figure 4: Instagram Explore / Search page with recommended content grid

4.4 Profile Module

Scenario 4a – View public business profile

Test Steps: Search for a business account (komobykaffa) → open profile → observe posts, followers, bio, contact, highlights.

Expected Result: Complete profile information displayed correctly.

Actual Result: Profile showed 220 posts, 32.6K followers, 4 following, the bio “Gujarat’s largest cafe now in Ahmedabad,” reservation contact number, opening hours and highlights.

Status: PASS

Screenshot – Business Profile (komobykaffa):

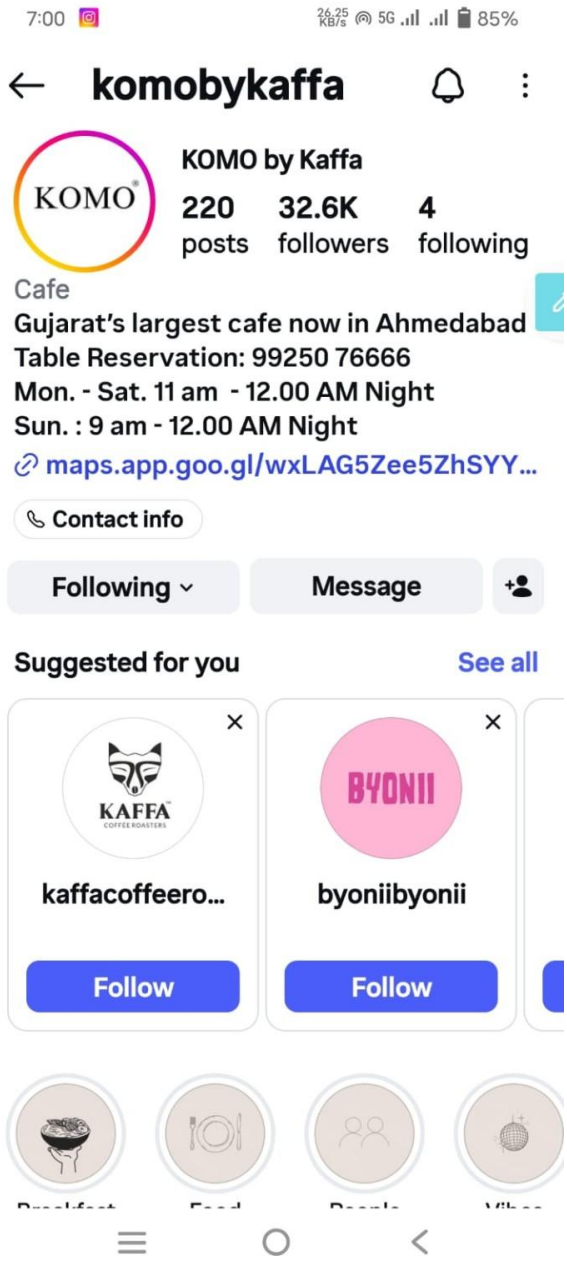


Figure 5: KOMO by Kaffa profile showing account statistics, bio, reservation information and highlights

4.5 Direct Messages (DM) Module

Scenario 5a – Open DM list and conversation

Test Steps: Open the Direct Messages tab → observe the conversation list and message previews.

Expected Result: Conversation list and individual threads load correctly.

Actual Result: Direct Messages list displayed conversations including Prakash Jha, Instagram, jitendarkatara, afsarali and others; unread indicators are visible.

Status: PASS

Screenshot – Direct Messages List:

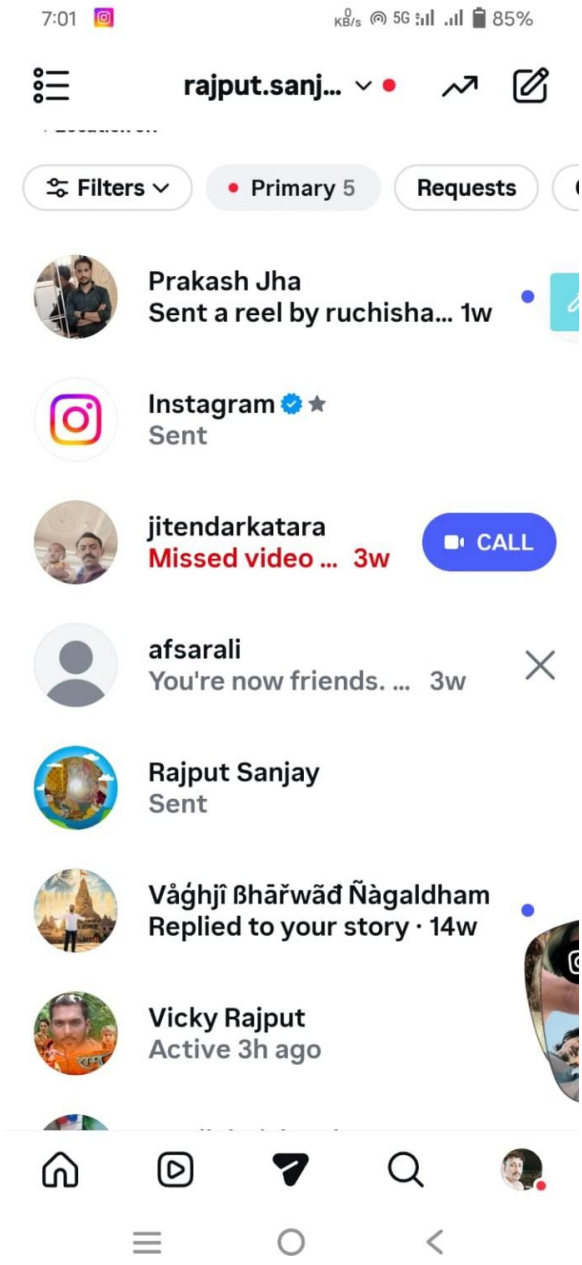


Figure 7: Instagram Direct Messages list showing conversations and message previews

4.6 Settings & Activity Module

Scenario 6a – Open Settings and check privacy

Test Steps: Open Profile → Menu → Settings and activity → observe the visible sections.

Expected Result: Settings and activity page opens and its available sections are visible.

Actual Result: Settings and activity page displayed Accounts Center, Saved, Archive, Your activity, Notifications, Time management, Instagram for tablets and Professional dashboard. Account privacy status is not visible in this screenshot.

Status: PASS

Screenshot – Settings and activity:

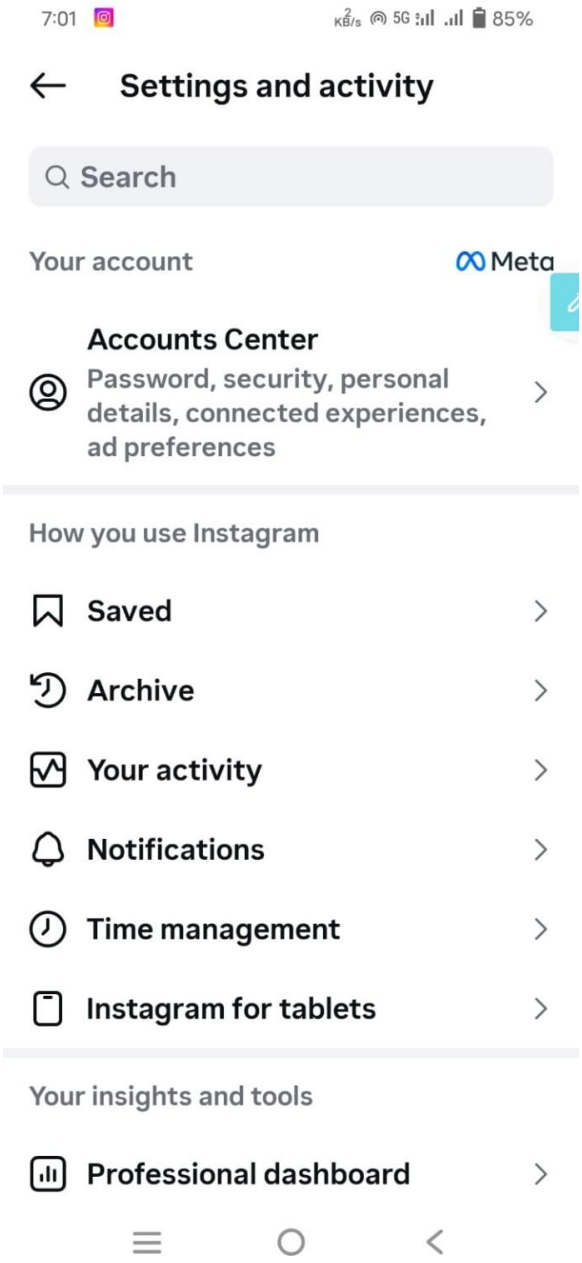


Figure 8: Instagram Settings and activity page showing Accounts Center and activity/settings options

5. Summary of Test Results

Screens Reviewed	Displayed	Issues Noted	Review Result
22	**22**	**0**	**100%**

6. Conclusion and Recommendations

The supplied screenshots show the Instagram login screen, Home Feed, post view, Explore grid, KOMO by Kaffa profile, Direct Messages list and Settings and activity page. The visible screens and controls can be reviewed through black-box and UI testing. Actions such as successful login, password reset, opening a DM thread and changing privacy settings were not demonstrated by the screenshots and should be executed separately before recording them as passed.

Recommendations: Continue regression testing after any major app update. Additional non-functional tests (performance under poor network, accessibility, security of session tokens) can be added in future practicals.

7. Bug Reports

No critical or major bugs can be confirmed from the supplied screenshots alone. The screenshots show the listed screens rendering, but further interaction testing is needed to verify functional behaviour.

— End of Report —